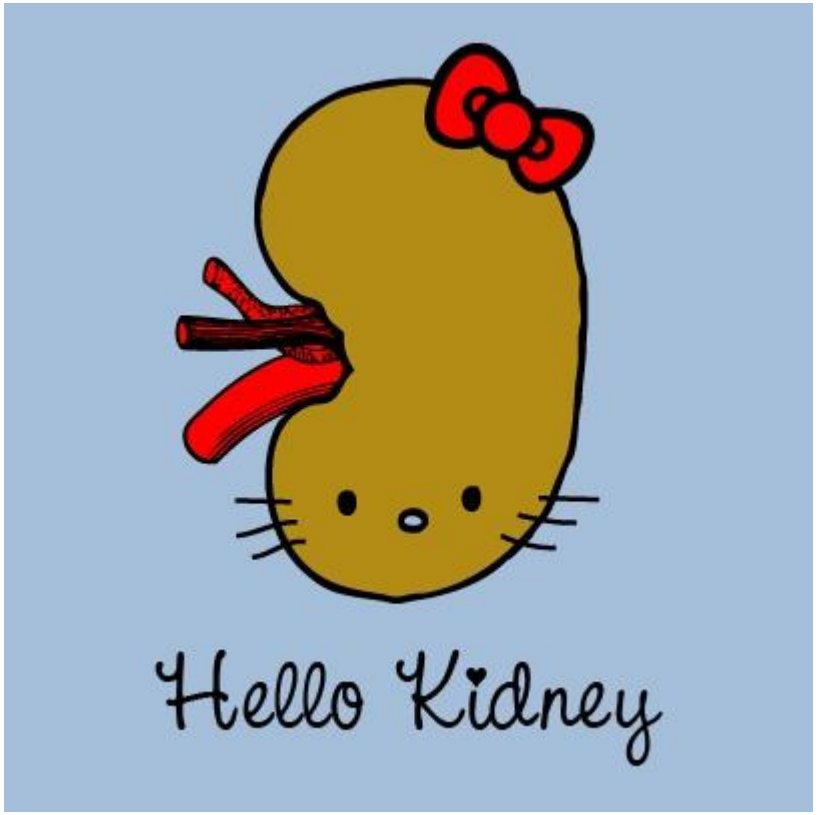
ACUTE NEPHRITIC SYNDROME



ACUTE POST-STREPTOCOCCAL GLOMERULONEPHRITIS

Definition

AGN that follows an infection with a *nephritogenic strain* of group A beta hemolytic streptococci.

The *classic example* of the acute nephritic syndrome.

Nelson Textbook of Pediatrics, 7th Edition

Epidemiology

- Most commonly – sporadic.
- Despite that, epidemics and clusters of cases - in some poor or rural communities

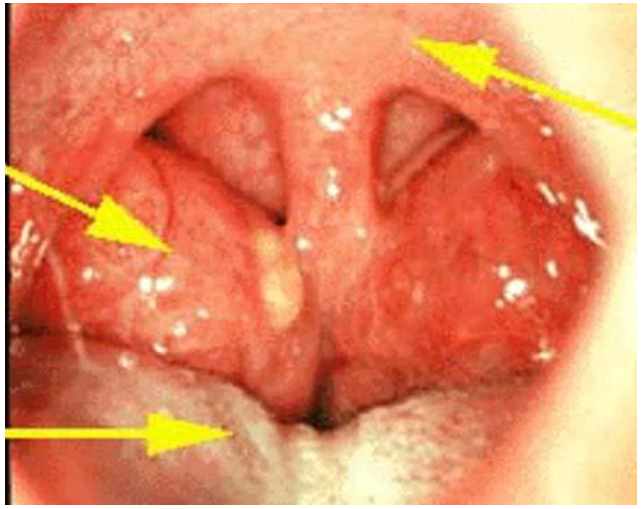
- Peak incidence - age 5-12 y/o.
- Male : female ratio is 2 : 1.

Etiology and Pathogenesis

- The child gets throat or skin infection by *nephritogenic strain* of group A beta hemolytic streptococci - serotype 12 , 4 and 1

- Antibodies to streptoccocus (eg antistreptolysin O) are formed in his circulation

- Antigen-antibody circulating immune complexes are subsequently deposited along the glomerular basement membrane (GBM).





Clinical Features

Typical Manifestation

Develop acute nephritic syndrome 1–2 wk after an antecedent streptococcal pharyngitis or 3–6 wk after a streptococcal pyoderma.

1. Edema

Face, periorbital area → lower extremities → generalized (ascites, pleural effusions)

2. Proteinuria :

3. Oliguria

Typical Manifestation (2)

3. **Gross hematuria**

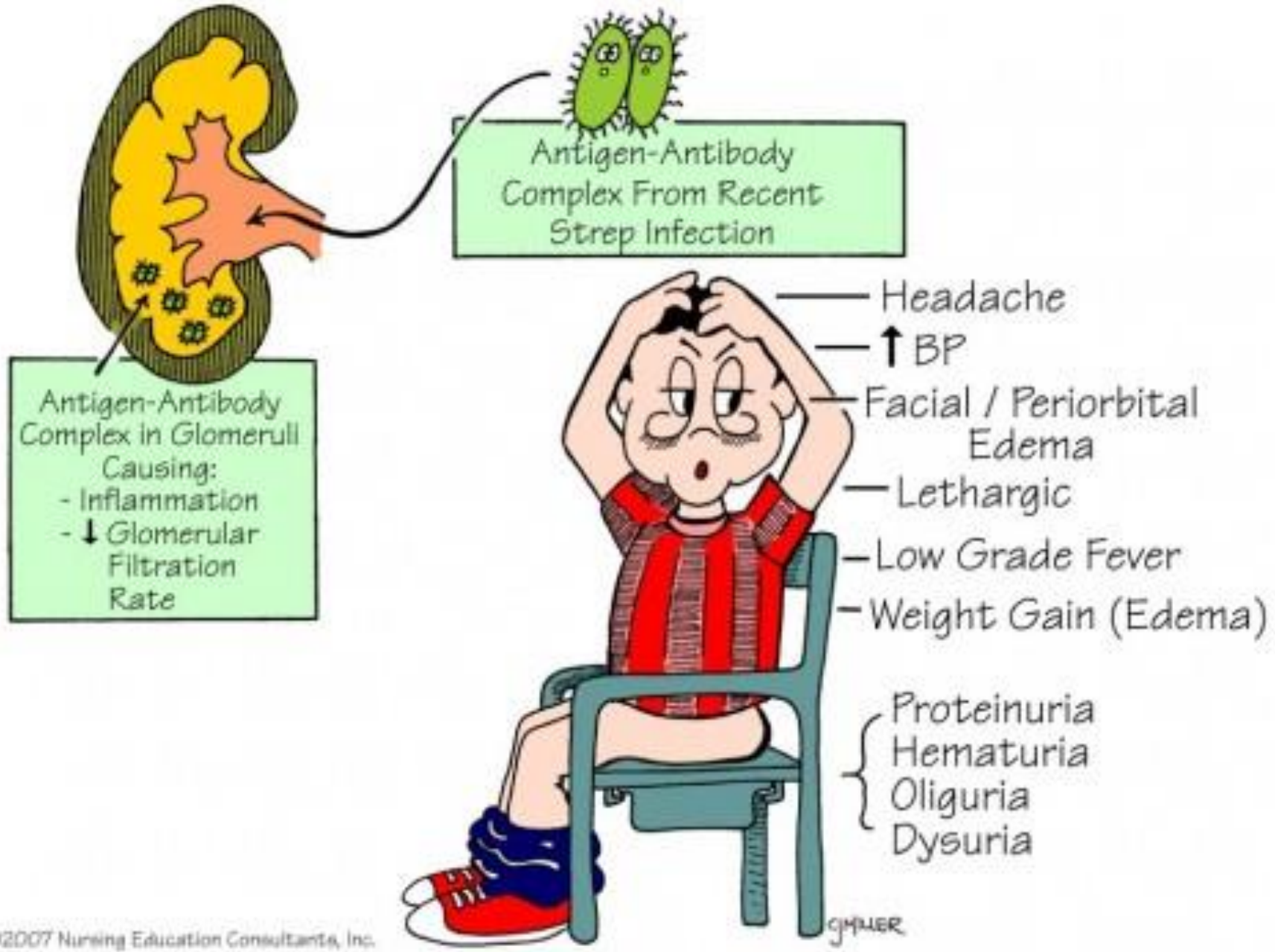
   Smoky, tea-colored, cola-colored, or fresh bloody urine

4. **Hypertension (50%) – mild to moderate, typically subsides promptly after diuresis**

5. **Nonspecific symptoms:**

   Such as anorexia, vomiting, general malaise, lethargy, abdominal or flank pain, low-grade fever and weight gain.

GLOMERULONEPHRITIS



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Complications in severe cases

- Circulatory hypervolemia / Congestive heart failure
- Encephalopathy
- Acute renal failure

Laboratory Investigations

1. Urinalysis
2. Bacteriological and Serological test
3. Renal function test
4. Full blood count
5. Serum complement levels
6. Kidney ultrasound

Laboratory Investigations

Urinalysis

- Macroscopic hematuria: Rusty or tea-color.
- Proteinuria 2+.
- RBCs casts.
- Pyuria.







Laboratory Investigations

Evidences of streptococcal infection

- Throat or skin cultures.
- Antistreptolysin O (ASO) titer: rising titre .
- Anti-deoxyribonuclease (DNase) B level.

Renal function Test: May be elevated

Hyperkalemia, hypocalcaemia, hyponatremia, and metabolic acidosis are seen only in severe patients.

C3: decreased

Diagnosis

- Acute onset
- Symptoms: edema, oliguria, dark urine, hypertension
- Urinalysis: RBCs, protein, casts
- Evidences of streptococcal infection:
  - Prodromes
  - Elavated serum titers of Abs to streptozymes(ASO)
- Serum C_3 - Reduced

- Treatment of APSGN is largely that of supportive care.
- Usually, patients undergo a spontaneous diuresis within 7 to 10 days after the onset of their illness - strict monitoring – nephrotic chart + fluid restriction until diuresis
- Management is directed at treating the acute effects of renal insufficiency and hypertension

Treatment

- Diuretics
- Diet

  Fluid restriction – during oliguric phase

  Sodium restriction is necessary: hypertension, heart faliure

  Protein restriction is unnecessary except if raised cheimstry.
- Antibiotics

  A **10-day course** of systemic antibiotic therapy with **penicillin** is recommended to limit the spread of the nephritogenic organisms.

  Antibiotic therapy does not affect the natural history of glomerulonephritis.

Antihypertensive drug used for hypertensive emergency in children	
Drug	Dose
Nifedipine	0.25 – 0.5 mg/kg/dose oral, may be repeated twice if no response. Caution with the sublingual route
Sodium nitroprusside	0.5 – 1.0 mcg/kg/min, iv infusion, may increased stepwise to 8.0 mcg/kg/min maximum. Needs to be given in the ICU setting . Caution: in liver and renal failure
Labetolol	0.2 – 1.0 mg/kg/dose by repeated iv boluses OR 0.25 – 2.0 mg/kg/hour by iv infusion.
Hydralazine	0.2 – 0.4 mg/kg/dose by iv bolus, may be repeated twice if no response.

Nephrotic Syndrome

Definition

Manifestation of glomerular disease, characterized by nephrotic range proteinuria and a triad of clinical findings associated with large urinary losses of protein ,hypoalbuminaemia , edema and hyperlipidemia

- Protein excretion of > 40 mg/m^2/hr
- First morning protein : creatinine ratio of > 2-3 : 1

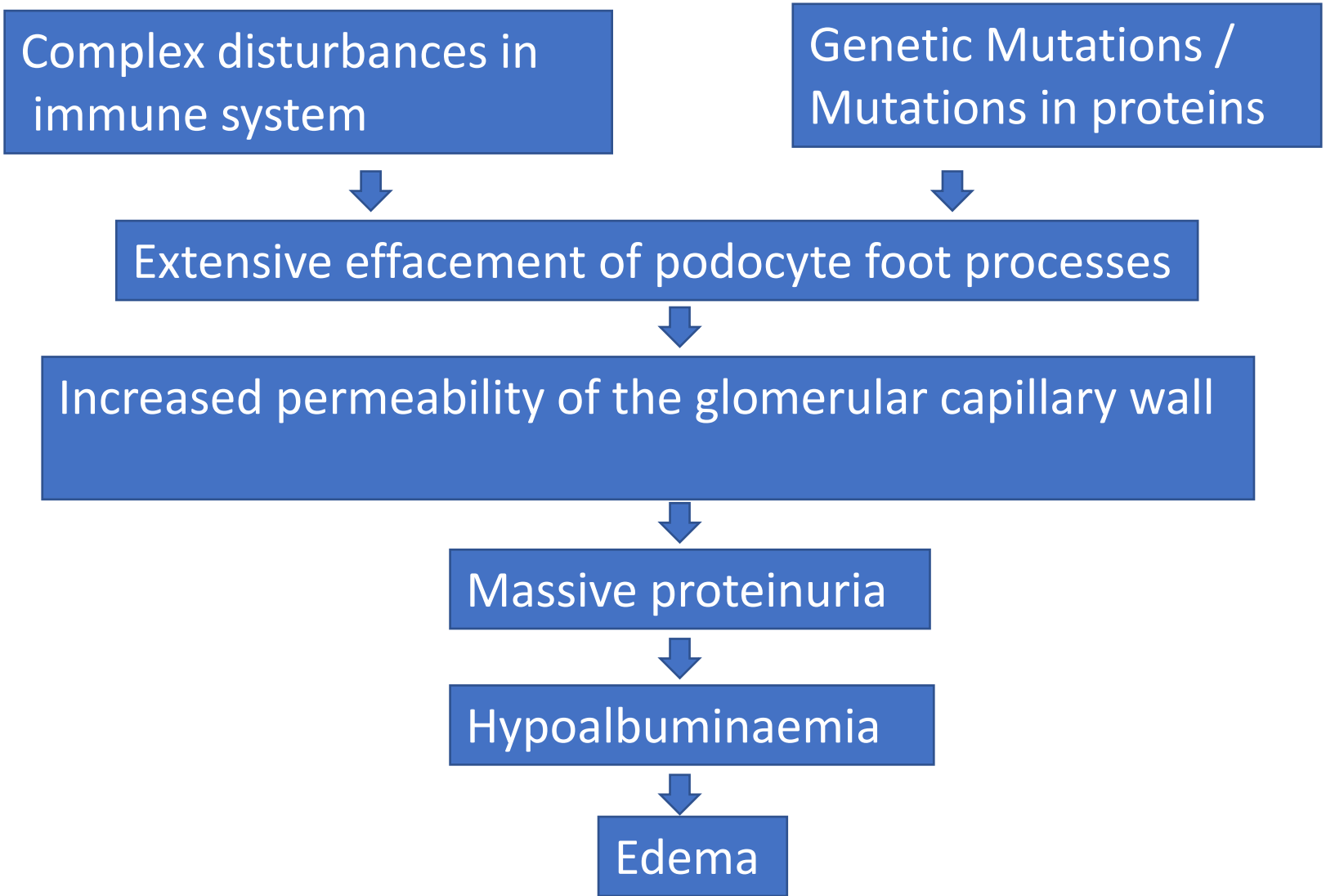
Incidence

- 2 – 7 cases per 100,000 children per year
- Higher in underdeveloped countries (South east Asia)
- Occurs at all ages but is most prevalent in children between the ages 1.5-6 years.
- It affects more boys than girls, 2:1 ratio

Etiology

- Genetic
- Secondary
- **Idiopathic** or Primary

❑ Pathophysiology



❑Clinical Features

- Edema

  Mild and start with peri orbital puffiness, lower extremities, then progress to generalized edema, ascites, pleural effusion, genital edema
- Decreased urine output
- Anorexia, Irritability, Abdominal pain and diarrhoea
- Absence of
  - Hypertension
  - Gross hematuria










Swollen face,
periorbital oedema

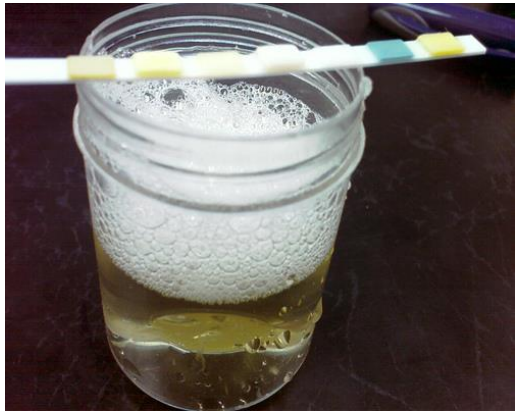
DIFFERENTIALS

- Protein losing enteropathy
- Hepatic failure
- Heart failure
- Acute/Chronic Glomerulonephritis
- Protein Malnutrition

Lab Investigations

- Urine Examination
- Complete Blood Count & Blood picture
- Renal parameters :
  - Spot Urine Protein : Creatinine ratio
  - Urinary protein excretion
- Liver Function Test
- Renal Biopsy ???

- Urinalysis - 3+ to 4+ proteinuria
- Renal Function
  - Spot UPC ratio > 2.0
  - UPE > 40 mg/m^2/hr
- Serum Creatinine – normal or elevated
- Serum albumin - < 2.5 gm/dl
- Serum Cholesterol/ TGA levels – elevated



Management

Initial Episode

- High protein diet
- Salt moderation
- Treatment of infections
- If significant edema :- diuretics & Aldosterone antagonist (Fursemide, spironolactone)
- Corticosteroid therapy with Prednisolone or prednisone

  (2mg/kg per day for 6 weeks followed by 1.5 mg/kg single morning dose on alternate days for 6 weeks)

DEFINITION FOR IDIOPATHIC NS

- **REMISSION:**

  Urinary protein excretion < 4 mg/m^2/hour or urine dipstick nil/trace for 3 consecutive days.

- **RELAPSE:**

  Urinary protein excretion > 40 mg/m^2/hour or urine dipstick ++ or more for 3 consecutive days.

Management of SRNS

Steroids + calcineurin inhibitors + ACE inhibitors + Diuretics

- The metabolism and excretion of many drugs and their pharmacologically active metabolites depend on normal renal function.
- Accumulation and toxicity can develop rapidly if dosages are not adjusted in patients with impaired renal function.
- A familiarity with basic pharmacologic principles and a systematic approach are necessary when adjusting drug dosages in patients with abnormal kidney function.

The distinct steps involve :

- calculating the patient's glomerular filtration rate
- Choosing and administering a loading dose
- Determining a maintenance dose
- Decision regarding monitoring of drug concentrations.

If done properly, therapy in renal patients should achieve the desired pharmacologic effects while avoiding drug toxicity.

- **An awareness of biologically active or toxic metabolites of parent compounds that accumulate when the glomerular filtration rate is reduced is also necessary to avoid toxicity.**
- **The effects of dialysis on drug elimination and the need for supplemental dosing are additional considerations in patients undergoing renal replacement therapy.**


Thank you
wiseGEEK